Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

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SAP ID:60017230102

ROLL NO.:A018

BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:19/04/24

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Pg-1

Java Exp-11

* Aim: To implement Abstract classes and Packages

* Description:-

1) Need of Abstract classes

→ Abstract classes is Needed because abstract class is a good choice if we are using the inheritance concept, since it provides a common base class implementation to derived class, An abstract class is also used if we want to declare non-public members. In an interface, all methods must be public and if you want to add new methods in the future, then an abstract class is a better choice.

2) Properties of abstract classes

→ i) An instance of an abstract class cannot be created

ii) An abstract class can contain constructors.

iii) We can have an abstract class without any abstract method

iv) Abstract classes can have final methods

v) In An Abstract class, we are not allowed to create an object.

vi) We can define static method in an abstract class.

vii) Abstract keyword can be used ~~to~~ for declaring top-level-classes as well as a inner classes as an abstract.

viii) If a class contain one abstract method, then compulsory that we should declare the class as abstract.

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3) Difference between Abstract Class and Interfaces

Abstract Class	Interface
1) It can have Abstract and non-Method	1) It can have only abstract method.
2) It does not support multiple inheritance.	2) It supports multiple inheritance.
3) It can have final, non-final static variables.	3) It can have only static and final variables
4) It can provide the implentation of interface.	4) It cannot provide the implementation of abstract class
5) The abstract keyword is used to declare abstract class	5) The interface is used to declare interfaces

4) Need and uses/benifits of packages.

→ Uses of packages are as follows:-

1) Packages organize the group of classes into a single API unit.
2) It will control the naming conflicts
3) The access protection will be easier. Protected and defualt are access level control to the package
4) Easy to locate the related classes.
5) Reuse the existing classes in packages.

Need of packages is to avoid name conflicts and to write a better maintainable code.

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1) b) WAP to create a package called volume having Cylinder class volume (). WAP that imports this package to calculate volume of cylinder.

Program :-

```
package Volume;
public class Cyclinder {
    double volume (double rad, int height) {
        double volcyc;
        double pi = 3.14;
        volcyc = pi * Math.pow(rad, 2) * height;
        return volcyc;
    }
}

import volume.Cyclinder;
import java.util.*;
class Main {
public static void main (String args [])
  {
    double radius, volcyl;
    int height;
    Scanner sc = new Scanner (System.in);
    System.out.println (" Enter the Radius : ");
    radius = sc.nextDouble();
    System.out.println (" Enter the height ");
    height = sc.nextInt();
    Cyclinder c = new Cyclinder();
    volcyl = c.volume (radius, height);
    System.out.println (" Volume of Cylinder = " +
```

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volyc1)g

}
}

Output:-

D:java > javac Main.java
D:java > java Main
D:java > Enter the Radius:
10
Enter the height:
10
Volume of Cylinder =3140

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Experiment No: 11

Aim : To implement Abstract classes and packages.

Program1 : Write an abstract class program to calculate area of circle, rectangle and triangle.

Code :-

```
import java.util.*;
abstract class Area{
abstract void calc(double dim1, double dim2);
void display(){
System.out.print("Area: ");
}
}
class Triangle extends Area{
double area;
void calc(double height, double base){
this.area = height*base/2;
}
void display(){
super.display();
```


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```
System.out.print(area);
}
}
class Rectangle extends Area{
double area;
void calc(double length, double breadth){
this.area = length*breadth;
}
void display(){
super.display();
System.out.print(area);
}
}
class Circle extends Area{
double area;
void calc(double radius, double pi){
pi = 3.14159;
this.area = pi*radius*radius;
}
void display(){
super.display();
System.out.print(area);
}
}
class AbstractArea{ public static void
main(String[] agrs){ Scanner input = new
Scanner(System.in);
```


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```
System.out.print("Enter base and height: ");
double base = input.nextDouble(); double
height = input.nextDouble();
Area a1 = new Triangle();
a1.calc(base, height);
a1.display();
System.out.print("\nEnter breadth and length: ");
double breadth = input.nextDouble(); double
length = input.nextDouble(); Area a2 = new
Rectangle(); a2.calc(breadth, length);
a2.display();
System.out.print("\nEnter radius: ");
double radius = input.nextDouble();
Area a3 = new Circle();
a3.calc(radius, radius);
a3.display();
input.close();
}
}
```


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Output :

```
Enter base and height: 25
2
Area: 25.0
Enter breadth and length: 25
2
Area: 50.0
Enter radius: 10
Area: 314.159
```


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Program2 : WAP to create a package called vol having Cylinder class and volume (). WAP that imports this package to calculate volume of a Cylinder.

Code :

```
package vol;
public class CylinderPackage{
double radius, height;
public CylinderPackage(double radius, double height){
this.radius = radius;
this.height = height;
}
public double volume(){
return Math.PI*radius*radius*height;
}
}
import vol.CylinderPackage;
```


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```
public class VolumePackageDemo{
public static void main(String[] args){
CylinderPackage cp = new CylinderPackage(10, 1);
double volume = cp.volume();
System.out.print("Volume: "+volume); }}
```

Output :

```
Volume: 314.1592653589793
```

Conclusion: Learnt To implement Abstract classes and packages

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